# LEC3 - Scales & Units



## Natural Units (Distance)

- Astronomical Unit (AU) ⇒ distance from earth to Sun ⇒ $1.5 \times 10^8$ km
- Light year (LY) ⇒ the distance light travels in 1 year ⇒ $9.5 \times 10^{15}$ km

## Natural Units (Speed)

- Kilometer per second (km/s) ⇒ 3600 m/s
- Speed of light (c) ⇒ $3 \times 10^8$ m/s

## Natural Units (Time)

- Year
- Mega Year ⇒ 1M years
- Giga Year ⇒ 1B years

Distance, speed, and time are mathematically related

Distance = Speed · Time

m = m/s · s

## Looking Back In Time

| Destination | Travel Time |
|---|---|
| Moon | 1 second |
| Sun | 8 minutes |
| Sirius | 8 years |
| Andromeda | 2.5M years |

⇒ We see objects as they appeared in the past, therefore the farther we look, the further we look back in time

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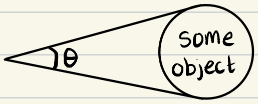
## Angles in Astronomy

**Angular Size:** A sizing classification based on how many degrees an object occupies in the sky

Often we don't know the distance to objects. The only thing we can measure is the angular size.

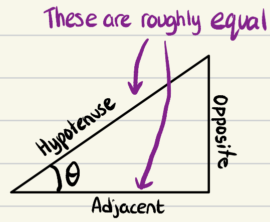



$\sin\theta = \frac{O}{H}$

$\cos\theta = \frac{A}{H}$

$\tan\theta = \frac{O}{A}$

You can practically use sin OR tan because due to the sheer distance to the object means adjacent and hypotenuse are roughly equal




**ANGULAR SIZE ≠ SIZE** ⇒ The Sun is roughly 400x bigger than the Moon, but it is roughly 400x farther. Therefore, their angular sizes are almost identical

## Natural Units

*We subdivide degrees into arcminutes, which we subdivide into arcseconds

Angles
- ↳ Degrees ($1^\circ$)
- ↳ Arcminute ($\frac{1}{60}^\circ$, $1'$)
- ↳ Arcsecond ($\frac{1}{60}'$, $1''$)

- Another way to classify object sizes is with angular size
- Angular size is determined by size AND distance, for example, the Sun and Moon have the same angular size (Think about solar eclipses)